

TECHNICAL ARSENAL & SKILLS

Artificial Intelligence & LLMs: Offline Hybrid RAG, Multi-Agent Systems (LangChain, LangGraph, LlamaIndex), Local SLMs (Ollama, llama.cpp), Vector Databases (Qdrant, Chroma, HNSW, BM25, LEANN Store), Cross-Encoder Reranking, LightRAG, Hugging Face Hub, MLOps.

Computer Vision & Edge Systems: YOLO (v8/v11), OpenCV, PaddleOCR, Edge AI Inference, Automated License Plate Recognition (ALPR), Biometric Face Verification & Anti-Spoofing, Real-Time Video Streams.

Backend & Full-Stack Development: Python, FastAPI, Flask, Node.js (TypeScript/JavaScript), React, PostgreSQL, Redis, Celery, Docker, Git, RESTful APIs, Linux Systems, Hardware VRAM & RAM Governors.

Engineering & Automation: Process Automation (n8n), Cyber-Physical Systems, Embedded Systems (C/C++, ESP32, Arduino), ROS, SolidWorks (CSWP), Classic Control & Hydraulics.

PROFESSIONAL EXPERIENCE & MILITARY SERVICE

- Air Force Engineer Soldier (National Service) — Egyptian Air Force**

Jan 2026 – Sep 2026

 - Served as Secretary to the Battalion Commander; performed technical engineering duties, handled executive administrative workflows, and supervised central-zone facility engineering and maintenance projects.
 - Engineered and deployed an internal point-of-sale (POS) and inventory management system for unit facilities, digitizing supply-chain tracking and daily sales records.
- Technical Researcher — Deputy Minister's Office for Institutional Development, MCIT**

Jul 2025 – Dec 2025

 - Evaluated AI-driven frameworks and streamlined digital workflows across 3+ cross-functional governmental committees, formulating technical reports for national digital transformation strategies.
- AI & Software Systems Engineer — RoboVAI Solutions**

2023 – Present

 - Authored robovai-ocr & Deployed "Roboty Parking AI":** Built and published the enterprise Python framework (`pip install robovai-ocr v3.0.0`) powering Smart Parking Operating Systems (SPOS); features multi-country ALPR (Egypt & GCC: KSA, UAE, Kuwait), automated barrier controls, 14-digit National ID parsing, and biometric anti-spoofing.
 - Engineered "RoboVAI Nova" AI Automation Platform (`nova.robovai.tech`):** Architected an enterprise AI business manager featuring autonomous multi-agent workflows, WhatsApp & e-commerce bots (Shopify, Salla, Zid), multi-platform social media scheduling, and integrated CRM lead pipelines.
 - Developed "RoboVAI Pro POS" Cloud Retail Platform (`pos.robovai.tech`):** Deployed a commercial multi-branch cashier and warehouse management system with real-time sales processing, dynamic receipt generation, stock alerts, and inventory ledger synchronization.
 - Architected "SecureAI-RAG" Enterprise Platform:** Engineered an on-premise sovereign document intelligence and offline RAG engine with zero cloud leakage, leveraging Qdrant, LangGraph supervisors, Celery, and real-time GPU VRAM governors (36/36 green E2E tests).
- Digital Transformation & AI Engineer — OMC Group**

Jun 2024 – May 2025

 - Implemented digital transformation across factory operations, HR, and supply chain departments, modernizing legacy enterprise workflows and eliminating data silos.
 - Integrated automated AI workflows (n8n, Python) across financial and operational departments, **reducing process bottlenecks by 30%.**
 - Designed and delivered hands-on AI literacy and practical automation training sessions for 50+ enterprise personnel, university students, and youth.
- Technical Operations Specialist — Merkato Industrial Platform**

Jan 2024 – Aug 2024

 - Streamlined B2B platform operations and end-to-end user onboarding workflows, boosting user retention and directly contributing to winning the **GESR SaaS Prize.**

EDUCATION

M.Sc. in Mechatronics Engineering — Arab Academy for Science, Technology & Maritime Transport (AASTMT)2026 – Present

Postgraduate Studies (Enrolled / In Progress) — Cairo, Egypt

B.Sc. in Mechatronics Engineering — High Institute of Engineering, 6th of October City2019 – 2024

Graduation Project (Grade: Excellent): *AI-Powered Environmental Robotics (Autonomous Oil Skimmer Boat)* — Developed an autonomous surface vessel integrating computer vision for oil slick detection, microcontroller logic, and IoT telemetry.

PRACTICAL FIELD TRAINING & INDUSTRIAL INTERNSHIPS

- Arab Organization for Industrialization (AOI):** Practical industrial training across the *Electronics Factory* (PCB/board assembly), *Arab American Vehicles (AAV)* (automotive assembly, chassis welding, and quality inspection), and *Renewable Energy Co.* (photovoltaic systems).
- Schneider Electric:** Technical training in industrial power distribution, control panel engineering, and automation solutions.
- El-Ekhlās Engineering Services & Manufacturing:** Practical training in industrial mechanical fabrication, precision CNC machining, and manufacturing workflows.
- IMT School:** Hands-on Embedded Systems training covering Embedded C, AVR/ARM microcontrollers, and hardware peripheral interfacing.
- Culture & Science City:** Completed 72-hour intensive summer program in *Hydraulic Systems & Classic Control*.

CERTIFICATIONS & CREDENTIALS

AI, Data & Software AI Specialist (80 Hrs, Score 85%): NTI & Huawei Talent Academy Generative AI Developer: Sprints, IOM UN & UK Dev Python & Digital Transformation: Microsoft & MCIT Cybersecurity Basics: Cisco Networking Academy	Hardware, Embedded & Robotics Arduino & IoT / Embedded: UC Irvine (Coursera) Arm Robotics (Grade: Excellent): Engineers Syndicate SolidWorks Design (CSWP Course): Udemy Digital Manufacturing & Design: SUNY Buffalo Industrial Assembly & Electronics: AOI (Arab Org. Industrialization)
Business, Marketing & Operations Digital Marketing Professional: IMTF (Portugal) Finance in Entrepreneurship: ICCD (Islamic Chamber) Digital Marketing MasterCourse: Udemy	Leadership, Governance & Training Professional TOT Diploma (40 Hrs): National Youth Council Sustainable Development Ambassador: Ministry of Planning Value Leadership Program: National Training Academy (NTA) Youth Innovation & Leadership: Friedrich Naumann & EYC

LEADERSHIP, COMMUNITY & PUBLIC IMPACT

- Engineering Syndicate (Giza Branch):** Active Member of the *Young Engineers Committee* and *Electronic Engineering Committee*; organizing technical programs, events, and developer workshops.
- Student Union Governance:** Former *Student Union President* (University / Institute); led the student body, managed campus initiatives, and coordinated with academic leadership.
- Community Initiatives:** Founder of *"Techno Egypt"* technical empowerment initiative; Active Member in *IEEE Computer Society (CS)* and *IEEE Robotics & Automation Society (RAS)*.
- Public Speaking & Media:** Featured tech expert on 5+ television networks (*Nile Cultural TV, TEN TV, Channel 2*); keynote speaker and workshop trainer at national tech conferences.
- Honors & Shields:** Awarded *Shields of Excellence* by the Libyan Ministry of Labor (Cybersecurity/IT) & Moroccan Chamber of Commerce. Official recognitions from MCIT, Ministry of Youth & Sports, and Police Academy.